

Algebra II with Trigonometry

Chapter 5.1

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Here is a comprehensive breakdown based on the provided document.

1. Document Summary

This document, titled "Chp 5.1: Angle Measure," introduces the foundational concepts of angles in trigonometry. It covers the definition of angles, positive/negative rotation, and standard position. It thoroughly explains the two main systems for measuring angles—degrees and radians—and how to convert between them. Finally, it applies radian measures to geometric and physical scenarios, specifically calculating the length of a circular arc, the area of a circular sector, and the linear and angular speeds of objects in circular motion.

2. Core Definitions, Theorems, and Formulas

Definitions:

- **Angle:** Formed by two rays with a common vertex. A rotation from an *initial side* to a *terminal side*.
- **Sign of an Angle:** Counterclockwise rotation is *positive*; clockwise rotation is *negative*.
- **Radian Measure:** In a circle of radius 1, the measure of an angle in radians is exactly the length of the arc that subtends the angle.
- **Standard Position:** An angle drawn in the xy -plane with its vertex at the origin and its initial side on the positive x -axis.
- **Coterminal Angles:** Two angles in standard position whose terminal sides coincide.

- **Linear Speed (v):** The rate at which the distance traveled along a circle is changing (distance divided by time).
- **Angular Speed (ω):** The rate at which the central angle is changing (number of radians divided by time).

Formulas:

- **Degrees to Radians Conversion:** Multiply the degree measure by $\frac{\pi}{180}$.
- **Radians to Degrees Conversion:** Multiply the radian measure by $\frac{180}{\pi}$.
- **Length of a Circular Arc (s):** $s = r\theta$ (where r is radius and θ must be in radians).
- **Area of a Circular Sector (A):** $A = \frac{1}{2}r^2\theta$ (where r is radius and θ must be in radians).
- **Linear Speed (v):** $v = \frac{s}{t}$
- **Angular Speed (ω):** $\omega = \frac{\theta}{t}$
- **Relationship Between Linear and Angular Speed:** $v = r\omega$

3. Worked Study Guide

Here is a step-by-step breakdown of how to think about and use these concepts.

Step 1: Understanding the Radian A radian isn't just an arbitrary number; it relates an angle directly to the radius of a circle. If you wrap the radius of a circle around its outer edge, that arc length subtends exactly 1 radian. Since the full circumference of a circle is $2\pi r$, a full revolution is 2π radians (which equals 360°).

Step 2: Converting Between Units Treat π radians and 180° as your conversion bridge.

- *If you have Degrees and want Radians:* You want to cancel degrees, so multiply by $\frac{\pi}{180^\circ}$.

(Example: $90^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{2}$ radians)

- *If you have Radians and want Degrees:* You want to introduce degrees, so multiply by $\frac{180^\circ}{\pi}$.

Step 3: Finding Coterminal Angles Because a full circle is 360° (or 2π radians), you can spin around the circle a full turn and end up in the exact same spot.

- To find coterminal angles in degrees: Add or subtract multiples of 360° .
- To find coterminal angles in radians: Add or subtract multiples of 2π .

Step 4: Using the Arc Length and Sector Area Formulas The formulas $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ are beautifully simple, but they carry a strict rule: **θ MUST be in radians**. If a problem gives you an angle like 40° , your very first step must be converting it to radians ($\frac{2\pi}{9}$ rad) before plugging it into $s = r\theta$ or $A = \frac{1}{2}r^2\theta$.

Step 5: Circular Motion (ω vs. v) Imagine a point on a spinning ceiling fan.

- *Angular speed* (ω) tells you how fast the angle is opening up (e.g., 10 radians per second). Every point on the fan blade has the *same* angular speed.
- *Linear speed* (v) tells you how fast a specific point is physically traveling through the air. A point on the tip of the blade travels a massive circle, so it moves very fast. A point near the center travels a tiny circle, so it moves slower. The formula $v = r\omega$ connects these ideas: linear speed scales up as the radius gets larger!

4. Practice Problems

Problem 1: Convert 150° to radians. *Answer:* Multiply by $\frac{\pi}{180}$.

$$150 \times \frac{\pi}{180} = \frac{150\pi}{180} = \frac{5\pi}{6} \text{ radians.}$$

Problem 2: Find one positive and one negative coterminal angle for $\frac{\pi}{4}$ rad.

Answer: Add and subtract 2π (which is $\frac{8\pi}{4}$). Positive: $\frac{\pi}{4} + \frac{8\pi}{4} = \frac{9\pi}{4}$.

Negative: $\frac{\pi}{4} - \frac{8\pi}{4} = -\frac{7\pi}{4}$.

Problem 3: A circle has a radius of 5 meters. Find the length of the arc that subtends a central angle of 1.5 radians. *Answer:* Use $s = r\theta$. Since the angle is already in radians, simply multiply. $s = 5 \times 1.5 = \mathbf{7.5 \text{ meters}}$.

Problem 4: Find the area of a circular sector if the radius is 6 cm and the central angle is 60° . *Answer:* First, convert 60° to radians: $60 \times \frac{\pi}{180} = \frac{\pi}{3}$. Next, use $A = \frac{1}{2}r^2\theta$. $A = \frac{1}{2}(6)^2(\frac{\pi}{3}) = \frac{1}{2}(36)(\frac{\pi}{3}) = 18(\frac{\pi}{3}) = 6\pi \text{ cm}^2$.

Problem 5: A wheel with a radius of 10 inches is spinning at an angular speed of 3 rad/sec. What is the linear speed of a point on the outer edge of the wheel? *Answer:* Use the relationship $v = r\omega$.

$v = 10 \text{ in} \times 3 \text{ rad/sec} = 30 \text{ in/sec}$. (Note: "radians" drop out as a unit when converting to linear distance).

5. Assumptions and Ambiguities Resolved

- **Radian Requirement in Formulas:** In Example 3(b), the document asks for arc length using a 40° central angle but does not explicitly print a warning that degrees cannot be used directly. I resolved this by emphasizing in the Study Guide that $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ are *only* valid when θ is in radians, which requires a conversion step first.
- **Typo in Example 7(b):** The text asks to find the linear speed of the blades in "in./m.". Given that part (a) calculates angular speed in "rad/min," it is highly assumed that "m." is a typo for "min." (minutes), not "meters," because the radius given is in inches. I assume the intended unit is "inches per minute."
- **Blank Examples:** The document consists of structured notes with blank spaces meant for the teacher (Ms. Welles) to fill in with students (e.g., Examples 3-7 have blank working spaces). I assumed the text of the examples was provided as context for the types of problems expected, so I created parallel problems for the Practice section rather than trying to solve the exact blank examples from the notes.